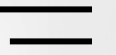


PCOS Risk Prediction and Lifestyle Recommendation System Using Machine Learning



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AGENDA

Literature Review

Methodology

Requirement Specification

Design Review (UI)

Design Review (DFD)

Design Review (ERD & Architecture)

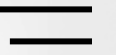




Literature Review

- PCOS is a widespread hormonal condition affecting many women of reproductive age and is linked with metabolic and lifestyle factors.
- Recent studies show that machine learning can support early risk screening using basic health indicators.
- Traditional diagnosis depends on clinical tests and doctor consultation, which may delay early awareness.
- Digital screening tools can improve accessibility and promote preventive care.

Methodology



- Identify key PCOS-related features such as BMI, symptoms, menstrual patterns, and lifestyle factors.
- Perform data preprocessing including cleaning, encoding categorical values, and scaling numerical inputs.
- Train machine learning models and compare their performance using standard evaluation metrics.
- Select the best-performing model for deployment.
- Integrate the trained model with a web interface for real-time prediction.





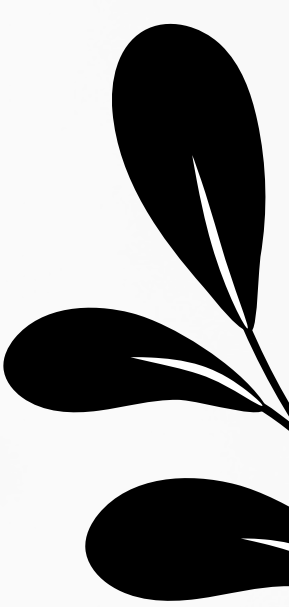
Requirement Specification

- The system should provide a simple web form for users to enter health and lifestyle details.
- It must generate PCOS risk prediction results quickly and reliably.
- The application should display key contributing factors and basic lifestyle recommendations.
- User inputs should be validated and handled securely without storing personal identifiers.
- The design should allow future enhancements such as chatbot integration.

Design Review (UI)

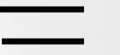
- Home page should clearly explain the purpose of the system.
- Assessment page should collect required inputs with minimal and simple fields.
- Results page should present risk level and recommendations clearly.
- A disclaimer section should communicate ethical and non-diagnostic usage.
- The interface should be responsive and readable on mobile and desktop devices.





Design Review (DFD)

- The user acts as the external entity interacting with the system.
- Input data flows from the user to the preprocessing module.
- The processed data is sent to the ML prediction module.
- Explainability logic interprets key contributing factors.
- Final results are displayed back to the user through the interface.



Thank You

